🏦 NPL Portfolio & Capital Strategy Manager

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Target Stakeholders: Special Servicing, Regulatory Governance, Investment Banking, Recovery Opertations & Strategy

Overview This specialized NPL Valuation Suite is a decision-support tool designed for Bank Recovery Managers and Credit Officers. It bridges the gap between distressed asset pricing (NPV) and regulatory capital efficiency (Basel III / IFRS 9) to determine the optimal disposition strategy for Non-Performing Loan tranches.

🚀 Key Features

- **Basel-Aligned Risk Grading:** Automatically assigns a 1A–4D Risk Grade based on Days Past Due (DPD) and Legal Recovery Friction.
  - **Numbers (1-4):** Represent PD Stages (Performing to Deep Distressed).
  - **Letters (A-D):** Represent LGD Levels (Low Friction to High Legal Cost).
- **Single Loan "Deep Dive":**
  - Dynamic NPV calculation with adjustable discount rates and legal "haircuts."
  - **Sensitivity Matrix:** A 4x4 heat map showing how sale prices shift across different regulatory grades.
- **Capital Release Simulator:**
  - Calculates the RWA relief gained from selling NPLs.
  - Demonstrates "Lending Multiplier" effect on new performing credit and shows how much new, performing credit can be issued by clearing "toxic" assets.
- **IFRS 9 Provisioning Adequacy Analysis:**
  - Compares Market NPV against Net Book Value to identify under-provisioned assets and quantify P&L impacts.
  - Visualizes Provisioning Adequacy to identify under-provisioned assets that require immediate attention.

- **Bulk Portfolio Processor:**
  - Generate randomized, regulatory-aligned test data or upload existing CSVs.
  - Provides a Total Portfolio Summary with weighted average yields and bolded totals for committee reporting.
  - Advanced CSV processing with a "Safety Net" mapping engine to handle diverse vendor data formats.

📊 Data Dictionary (Bulk Upload Template)

To use the Bulk Portfolio Upload feature, ensure your CSV contains the following fields. The tool's mapping engine will automatically handle minor variations in header names (e.g., "Days Past Due" vs "DPD").

Column	Data Type	Description
Loan_ID	String	The unique identifier for the loan or borrower.
Gross_Recovery	Numeric	The total estimated amount expected to be recovered (Face Value).
DPD	Integer	Days Past Due. This drives the **Basel PD Grade (1-4)**.
Years	Integer	Expected timeline to reach full liquidation/settlement.
Rate	Numeric	The Target Discount Rate % (Cost of Capital/Buyer's ROI).
Legal_Costs_Pct	Numeric	The % "haircut" for legal/servicing fees. Drives the **LGD Grade (A-D)**.
Collateral	Option	Must be exactly **"Secured"** or **"Unsecured"** to calculate the Capital Charge correctly.

Bulk Upload Template file link:
https://github.com/gram99/NPL_Valuation_Strategy/blob/main/NPL.csv

When using the bulk upload template, do not remove the column names in the first row of the template. All data from row two down on the template may be removed. If the first row column heading names are removed inadvertently, ploease use the data listed under the **Column Heading** in the table shown above.

🛠️ Technical Stack

- **Language:** Python 3.x
- **Interface:** Streamlit
- **Data Handling:** Pandas & NumPy
- **Visualization:** Altair (Provisioning Adequacy Charts)

📈 Managerial Value Proposition

"The goal isn't just to calculate a price; it's to optimize the balance sheet. By identifying 4D assets early and quantifying the capital release potential, managers can justify immediate sales that free up liquidity for higher-yielding, performing loans."

🧠 Best Practices: Stage Migration & Strategy

The tool's Basel Risk Grade serves as a signal for IFRS 9 Stage Migration. Use the following guidelines to align the tool's output with your bank's provisioning policy:

1. **Stage 2 Migration (DPD 31-90 | Grade 2)**

- **The Signal:** The tool will shift from Grade 1 to Grade 2. This indicates a "Significant Increase in Credit Risk" (SICR).
- **The Strategy:** Priority should be Consensual Workout. Use the Provisions Analysis tab to check if the current NPV is still above the Book Value. If NPV begins to drop, it is often cheaper to settle now than to wait for Stage 3 litigation.

2. **Stage 3 Entry (DPD 91+ | Grade 3)**

- **The Signal:** The tool shifts to Grade 3. The asset is now officially "Non-Performing."

- **The Strategy:** Transition from Workout to Legal Recovery. If the Legal Friction (LGD) is Grade C or D, the Capital Release Simulator will likely show that an immediate sale is more capital-efficient than a multi-year foreclosure.

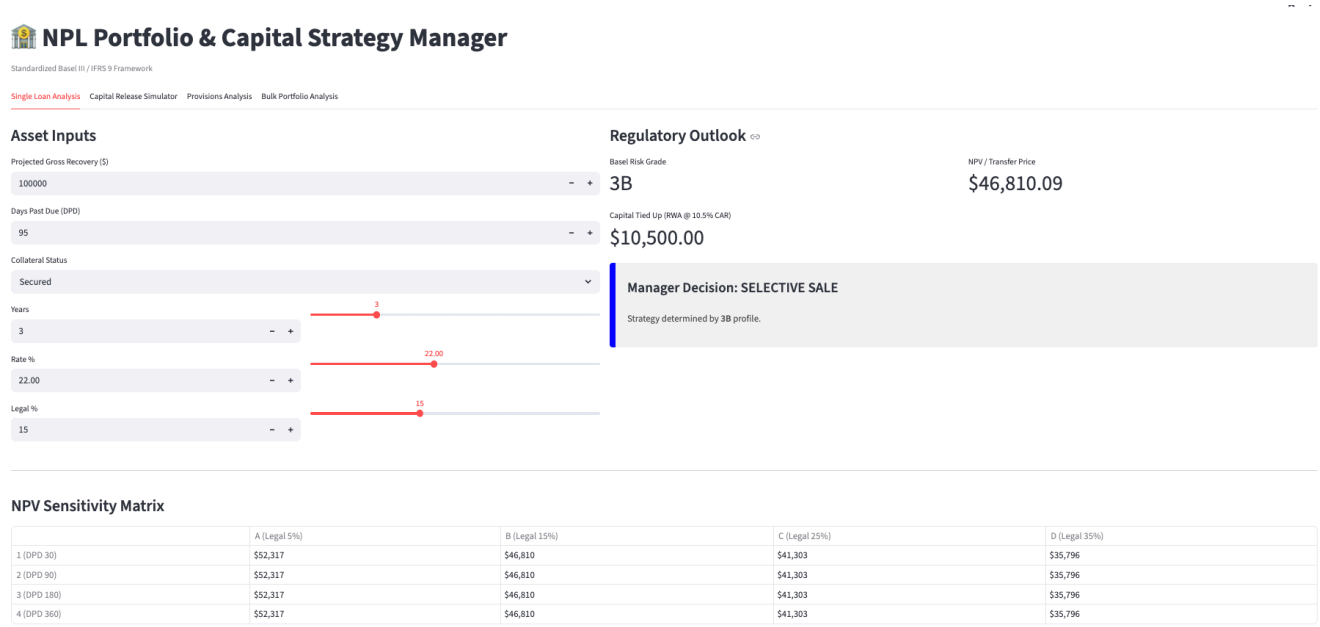
3. **Default & Loss (DPD 180+ | Grade 4)**

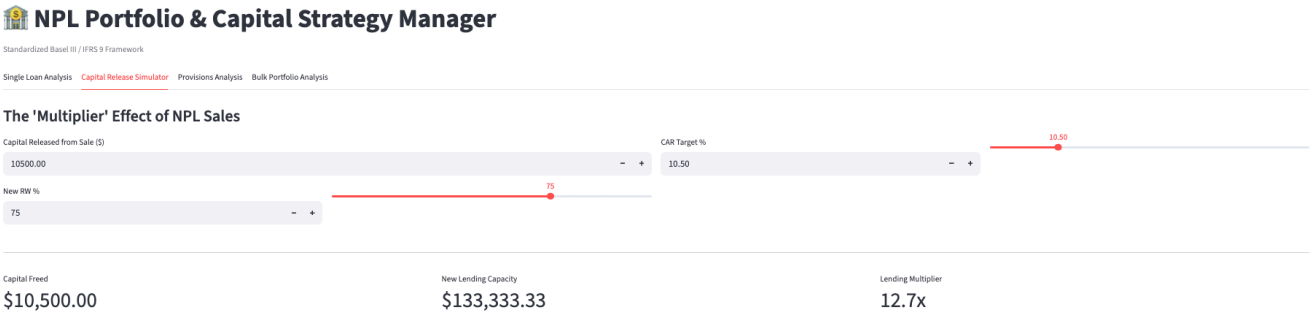
- **The Signal:** The tool shifts to Grade 4.
- **The Strategy:** Immediate Disposition. At this stage, the capital charge (RWA) is at its peak (up to 150%). Even a deep-discount sale often results in a "Capital Gain" when you factor in the Lending Multiplier created by freeing up those reserves.

4. **Managing the "Provisioning Gap"**

- Use the Provisioning Adequacy Chart monthly. If the Orange Bar (Book Value) remains significantly higher than the Blue Bar (Market NPV), you are carrying "latent risk.
- **"Pro Tip:** Align your Discount Rate in the tool with recent market benchmarks from NPL secondary markets to ensure your "Gap" analysis is realistic.

Screenshots





NPL Portfolio & Capital Strategy Manager

Standardized Basel III / IFRS 9 Framework

Single Loan Analysis | Capital Release Simulator | Provisions Analysis | Bulk Portfolio Analysis

IFRS 9 Provisioning & Adequacy

Existing Provision ($)

35000.00

Net Book Value

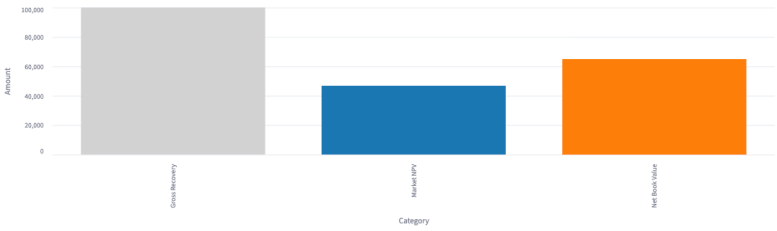
$65,000.00

P&L Impact of Sale

$-18,189.91

-18189.914574347633

Metric	Amount
Gross Recovery	$100,000.00
Existing Provision	$35,000.00
Net Book Value	$65,000.00
Market NPV	$46,810.09



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Standardized Basel III / IFRS 9 Framework

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Portfolio Processing

Generate & Load Sample Portfolio

Portfolio Gross

$2,800,000.00

Portfolio NPV

$1,143,404.32

Weighted Avg Yield

40.8%

Loan_ID	Gross_Recovery	DPD	Years	Rate	Legal_Costs_Pct	Collateral	NPV	Grade	Strategy
L-1000	$500,000.00	362.000000	3.000000	33.000000	18.000000	Unsecured	$174,272.53	4C	SELECTIVE SALE
L-1001	$100,000.00	64.000000	3.000000	24.000000	33.000000	Secured	$35,140.65	2D	IMMEDIATE SALE
L-1002	$50,000.00	335.000000	2.000000	26.000000	20.000000	Secured	$25,195.26	4C	SELECTIVE SALE
L-1003	$50,000.00	300.000000	5.000000	17.000000	21.000000	Unsecured	$18,016.39	4C	SELECTIVE SALE
L-1004	$50,000.00	60.000000	4.000000	27.000000	34.000000	Unsecured	$12,685.25	2D	IMMEDIATE SALE
L-1005	$500,000.00	95.000000	2.000000	31.000000	9.000000	Unsecured	$265,136.06	3B	SELECTIVE SALE
L-1006	$500,000.00	322.000000	3.000000	29.000000	34.000000	Secured	$153,725.11	4D	IMMEDIATE SALE
L-1007	$100,000.00	288.000000	5.000000	30.000000	5.000000	Unsecured	$25,586.26	4A	IMMEDIATE SALE
L-1008	$50,000.00	134.000000	5.000000	31.000000	31.000000	Secured	$8,942.58	3D	IMMEDIATE SALE
L-1009	$100,000.00	8.000000	2.000000	26.000000	34.000000	Unsecured	$41,572.18	1D	IMMEDIATE SALE

Upload Vendor CSV

Upload 200MB per file • CSV